

PRESENT DATA SCI-DRIVE SOLUTION



Smart AI Photo Cleaner APP

A Fast, Private, and AI-Powered Photo Management App

Group 15

Chia-Wei Chang (101570243)

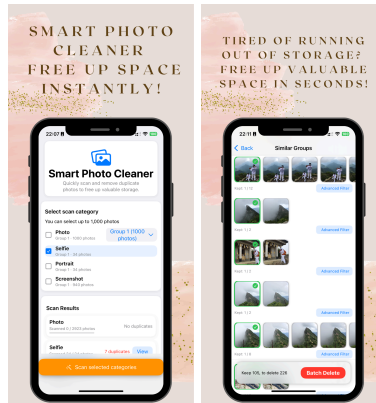
Applied A.I. Solutions Development Program

Table of Contents

1. Executive Summary.....	2
2. Assumptions and Context.....	2
3. Problem Statement and Opportunity.....	2
4. Market Analysis.....	2
5. Proposed Solution.....	2
6. What it is, what it does, and how it addresses the problem.....	2
7. Build vs Buy analysis.....	2
8. Financial Overview.....	2
9. Cost estimation, revenue potential, ROI.....	2
10. Stakeholder Analysis.....	2
11. Risk Analysis.....	2
12. Risk, impact level, likelihood, mitigation strategy.....	2
14. Implementation Plan.....	2
15. Timeline, milestones, team structure.....	2
16. Conclusion and Recommendations.....	3

1. Executive Summary

Smart AI Photo Cleaner Application, the goal is to help users clean up duplicate and unnecessary photos on their phones. The app will use lightweight models with 25 million parameters to detect similar or identical images and delete the rest by user selection. It will also classify images. Such as screenshots, so users can decide whether to delete them. Lastly, everything will run locally on the device for privacy and efficiency.



2. Assumptions and Context

- Users are overwhelmed by too many similar photos from burst shots.
- Manual cleanup is time-consuming and difficult.
- Most users have limited storage, which fills up fast.
- Cloud-based apps raise privacy concerns.
- On-device AI can provide a secure, fast alternative.

3. Problem Statement and Opportunity

Modern phones take many similar high-resolution photos, quickly consuming storage. Deleting these manually is a pain. There's a growing need for an app that can identify and remove duplicates easily and safely, without uploading personal photos to the cloud.

4. Market Analysis

- Most photo cleaner apps are paid or process data in the cloud.
- Competitor pricing: \$29.99 – \$39.99
- High demand for privacy-friendly, free alternatives.

5. Proposed Solution

Smart AI Photo Cleaner uses CoreML ResNet-50 to create embeddings of photos and compare them using cosine similarity. It groups similar images and lets users clean them with one tap. All processing happens locally on the device, so no photos are ever uploaded.

6. What it is, what it does, and how it addresses the problem

- Finds duplicate and similar photos instantly
- Categorizes photos (Selfies and Screenshots, etc.)
- Let users keep or delete with a swipe
- Protects privacy by doing everything on-device
- Helps free up storage and organize photos effortlessly

7. Build vs Buy analysis

- Build:
 - Quick (launched in 4 days)
 - Free Apple CoreML model with MIT license
 - No cost for cloud services
- Buy:
 - Higher cost (commercial solutions \$30–40)
 - May require API subscriptions and data hosting
- Verdict: Building is faster, cheaper, and more private.

8. Financial Overview

- Development Time: 4 days
- Tools: Free open-source + Xcode
- Monetization: Google Ads (next iteration)

9. Cost estimation, revenue potential, ROI

- Development Cost: Very low (solo developer, no backend)
- Revenue Potential: \$100+/month from ~5,000 active users via ads
- ROI: High, considering near-zero cost and ongoing passive income

10. Stakeholder Analysis

- Users: Benefit from better storage and privacy
- Developers: Low effort, scalable
- Advertisers: Gain visibility through in-app ads
- App Stores: Feature privacy-first utilities

11. Risk Analysis

- Risks
 - AI misclassifies photos
 - Slower performance on older devices

- Users fear losing important images
- Mitigation
 - Keep improving AI model accuracy
 - Add “undo” or “recently deleted” option
 - Optimize for older phones

12. Risk, impact level, likelihood, mitigation strategy

Risk	Impact	Likelihood	Mitigation
Misclassified Photos	High	Medium	Better model
Slow Scanning on Old Phones	Medium	Medium	Optimize processing
Accidental Deletion	High	Low	Add recovery option/preview

13. Implementation Plan

- Day 1: App idea, UX design, ResNet integration
- Day 2: Scanning logic, similarity detection
- Day 3: UI for grouping, preview, swipe actions
- Day 4: Final polish, App Store testing, launch

14. Timeline, milestones, team structure

Milestone	Day
Design & Setup	Day 1
AI + CoreML Setup	Day 2
UI Development	Day 3
Launch & Monetization	Day 4

15. Conclusion and Recommendations

Smart AI Photo Cleaner offers a fast, secure, and user-friendly way to manage photo storage. It was built quickly with free tools and offers strong financial potential. Future improvements should include better AI and more categories to enhance trust and performance.